

UNIVERSITI MALAYA

KIE4002 Final Year Project

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Proposal Defence

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FYP Title:	Simultaneous Localization of Multiple Partial Discharge Sources
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Objective

1. Digital Twin Simulation

Develop a validated digital twin environment of EM wave propagation during multiple partial discharges (PD) using FDTD/FIT simulators. This must incorporate experimental data from at least three distinct PD sources (e.g., corona, surface, void) and be considered complete upon achieving a shape similar to experimentally measured data by the end of semester 1.

2. Clustering of PD Sources

Evaluate and compare the performance of at least three distinct feature extraction methods (e.g., wavelet analysis, envelop detection, statistical parameters) and three separation/clustering techniques (e.g., k-means, DBSCAN, two-step algorithm) using the collected/simulated EM signal dataset. The goal is to successfully cluster the PD sources for the distinct PD types investigated in Goal 1.

3. Innovation and Optimization

Develop and implement a novel separation/clustering method by modifying or combining existing techniques (e.g., a hybrid clustering approach). This novel method must demonstrate a quantifiable improvement over the best-performing technique from Goal 2, and at the very least achieve a target clustering accuracy of 90% or higher in clustering ≥ 2 different PD sources.

Problem Statement

Partial Discharge (PD) is a localized electrical discharge that partially bridges the insulation between conductors, due to imperfections in the insulation. Under high voltage stress, the electric field in the defect exceeds a critical threshold, causing a discharge. Over time, this damage degrades the insulation, causing void expansion and chemical breakdown, ultimately leading to complete insulation breakdown (Sharma & Bhanddakkar, 2015).

Traditional methods of PD detection use voltage measuring devices across an impedance to detect a current spike in the device. This process is expensive and limited to devices electrically connected to the measuring instrument. A newer technology is to use Ultra-High Frequency (UHF) sensors to detect electromagnetic (EM) waves emitted from the discharges. Utilising the time difference of arrival (TDOA) of the signals at different receivers, the PD location can be determined (Chan et al., 2025). Multiple optimization methods to localize PD from the TDOA have been developed. However, many of these methods only apply to localizing one source. In practice, there will always be multiple PD sources discharging repeatedly, overlapping, attenuating, refracting, reflecting and taking multiple paths as they travel through obstacles in a substation to reach the receiver. Signals often overlap, and higher amplitude signals can mask lower amplitude ones. Therefore, a separation algorithm is needed to cluster incoming signals according to their sources for UHF radiometric detection, and to separate mixed signals successfully.

Furthermore, obtaining real world PD data for development purposes pose many challenges. First, real world data would contain interference from locally transmitted radio, as well as signal attenuation (Upton et al., 2021). This would require knowledge on signal denoising for proper development to be done, raising the bar of entry into the field. In addition, accurately emulating PD would require a lot of resources to fabricate the PD emulator and obtain HV test equipment. Even so, emulators may not represent a wide range of possible PD types that can occur. Obtaining access to enter a substation to do testing also proves to be difficult to do, requiring permission from the relevant authorities. Therefore, a method to accurately simulate UHF signals obtained from PD sources in different environments needs to be developed, allowing for separation algorithms to be tested preliminarily before deploying them to real-world scenarios. In fact, this field has been at the forefront of PD research for many years (Loubani et al., 2019).

Literature Review

The detection, and localization of PD have been under research since 1898 (Stone et al., 2022). Traditionally, according to IEC 60270, PD would be measured using a coupling capacitor in parallel with the device under test (DUT) to supply charge for the current spike. Using an impedance to convert the current spike into output voltage, it is measured via a suitable coupling device (CD) and measuring instrument (MI). A high-frequency current transformer (HFCT) can also be used to obtain a voltage signal of the current spike. Non-contact methods, which have been recently popularized in research, such as ultra-high frequency (UHF) and acoustic methods are introduced in IEC 62478. There are several methods to localize a PD source from its emitted UHF waveforms. Time difference of arrival (TDOA) utilizes the difference in time between arriving signals at different receivers to estimate the distance of the PD from each of them. The exact moment of signal arrival is usually determined from the first peak or cumulative energy plot. A set of equations that represent the relationship between TDOA and PD location is solved using optimization algorithms, such as Newton-Raphson, particle swarm optimization and ant colony optimization (Chan et al., 2025). The direction of arrival (DOA) method uses phased antenna arrays to determine which direction the PD is coming from, while the received signal strength (RSS) method uses the principle that the strength of received signal corresponds to the distance travelled by signal (Chen, 2019) (Upton et al., 2021).

When multiple PD sources are repeatedly discharging and emitting signals, it becomes difficult to differentiate each received waveform. Multiple methods to separate and cluster signals obtained from the voltage output waveform obtained from IEC 60270/HFCT measurements have been developed. Typically, features are extracted from the signals in the time, frequency or time-frequency domain, and a clustering algorithm is then applied. The signals arriving at a sensor can either be overlapped or non-overlapped. The non-overlapped case is simpler. One example of a developed method for that case is wavelet decomposition (WD) + principal component analysis (PCA) + density-based spatial clustering of applications with noise (DBSCAN). WD passes the signal through a series of complementary high pass and low pass

filters, to decompose it into detail and approximate coefficients. The signal can then be represented by scaled and shifted versions of the Mother Wavelet, a basis function defined by the user. PCA reduces the dimensionality of the sources to 3 and DBSCAN uses density to form clusters (Hao et al., 2011). Another method is to use simple geometrical structures to form envelopes of the waveform and create a time-frequency sparsity map (J. C. Chan et al., 2015). Similar methods can be found in (Wang et al., 2015), (Nimmo et al., 2017) and (Zhu et al., 2017). The overlapped case is more complex and requires blind source separation (BSS), a process to disentangle mixed signals (Tang et al., 2009). Research has shown that wavelet analysis and envelope comparison can also be used to separate sources from UHF EM waveforms (Yang & Judd, 2003) (Pinpart & Judd, 2010).

Machine learning has also been explored and applied to the field of PD clustering and separation. An application of time-frequency analysis methods, like scatter wavelet transform (SWT) and Hilbert-Huang transform (HHT) to extract PD features, and deep learning (DL) models to cluster them is detailed by Almeshdhar & Prochazka (2025). They achieved 100% experimental accuracy in PD separation. Carvalho et al. (2024) experimented with machine learning (ML) models (K-means, gaussian mixture model, mean-shift) to cluster PD according to features such as crest factor, skewness and kurtosis, achieving 93% accuracy. Further research on ML for PD clustering is stated in (Fu et al., 2023), (Zhang et al., 2022) and (Rauscher et al., 2024).

Most attempts at creating a digital twin of PD occurrence, represent partial discharge using a gaussian pulse of varying physical lengths, amplitudes and pulse widths, in software such as XFDTD or CST Microwave Studio. The software can numerically solve the E and H fields surrounding the PD source using the principles of finite-difference time domain (FDTD) or finite integration technique (FIT). Attempts at simulating PD are given in (Zanjani et al., 2012), (Jaber et al., 2016), (Ohtsuka et al., 2006), (Hoshino et al., 2008), (Umemoto & Tenbohlen, 2018), (Zheng et al., 2023), (Tian & Kawada, 2011), (Loubani et al., 2019), (Yun et al., 2020) and (He et al., 2020). A notable exception would be (Zheng et al., 2023), which does not use a gaussian pulse, but rather models the current pulse using the dielectric barrier discharge (DBD) model. Most of the simulations were done with the purpose of validating the accuracy of simulated single PD waveforms, compared to experimental measured waveforms. Only (Yun et al., 2020) and (He et al., 2020) investigated the simulation of multiple PD sources for the purpose of applying separation/clustering algorithms to them. He et al. (2020) implement a method to separate gaussian-pulse simulated PD signals from signal bispectra characteristics. The algorithm is then experimentally validated in a substation. More research would be needed to confirm the eligibility of testing separation algorithms on simulated PD waveforms, on whether a simulation would be capable of producing different types of PD emitted waveforms that accurately resemble experimentally collected samples, with extractable features that are unique to each different simulated PD.

Proposed Methodology

1. Digital Twin Simulation Method

To achieve the first objective, a digital twin environment would be created. Two different methods of modelling a PD occurrence will be explored. The first is to use a gaussian current pulse and simulate the EM waves propagating from it:

$$I(t) = I_{max}e^{-t^2/2\sigma^2}$$

Where I_{max} is maximum current amplitude, σ is the pulse width (Loubani et al., 2019).

The second would be to use the dielectric barrier discharge model to compute the current pulse waveform (Zheng et al., 2023).

To simulate the propagation of EM waves from the PD source, two simulation software have been identified. The first would be CST Microwave Studio, which uses the finite integral technique (FIT), a close relative of FDTD to model EM wave propagation. It is simpler to use because of its GUI but the educational license lacks capability. The second would be an open-source software called OpenEMS, which uses FDTD and has a steeper learning curve due to its command line nature, but would have more computing power and options.

The FDTD method involves solving Maxwell's equations in the time domain using finite-difference approximations. It discretizes space into Yee cells, with E and H components computed at the edges.

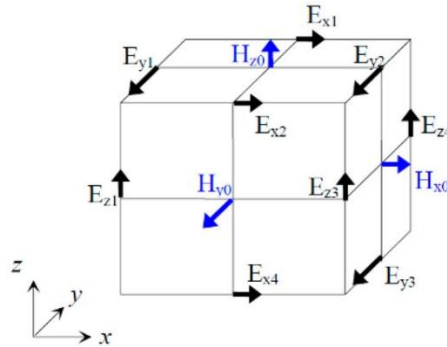


Figure 1: A Yee cell (Loubani et al., 2019).

Time is discretized using time steps, with E and H components updated at each time step.

2. Feature Extraction Methods

Four methods to extract features from the simulated/experimental PD UHF waveform have been identified: wavelet analysis (Yang & Judd, 2003), envelope detection (Pinpart & Judd, 2010), selected bispectra (He et al., 2020), and statistical analysis (skewness, kurtosis, crest factor, etc). These four would be implemented, and their performances compared. The algorithms would be implemented in Jupyter notebook for simplicity. Innovations such as

applying a method not created specifically for UHF waveforms, or modifying existing methods will be experimented with.

3. Clustering/Separation Methods

Several different clustering/separation methods have been identified during the literature review, such as the K-means algorithm, two-step algorithm, DBSCAN, and deep learning (DL). Algorithms will be selected for implementation and comparison based on their proven success rate and suitability. Again, innovations mentioned above would be carried out. In addition, if time permits, an attempt to create a neural network for PD clustering will be made. Google Colab will be used for the model construction and training.

4. Experimental Validation

Planar monopole antennas and a digital oscilloscope will be used to collect EM waves. Multiple types of PD sources will be emulated using piezoelectric discharge sources. If feasible, other types of PD emulators will be obtained or constructed, such as a point plane electrode or insulation defect model, shown in the figures below.



Figure 2: Insulation defect model (left) and point plane electrode (right) (Li et al., 2023).

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